DATA SCIENCE & ARTIFICIAL INTELLIGENCE



Probability

Lecture No. 05



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Recap of previous lecture



Topic

Probability (Part 04)

[Conditional Probability]

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Topics to be Covered



Topic

BASICS of PROBABILITY (Part-05)

(Baye's Theorem)

Q (2015) Parcels are sending from sender S to Receiver R sequentially through two post offices. The prob of loosing an incoming parcel is $\frac{1}{5}$ by each post office. independently of all other parcels. Given that Parcel has lost then find the Prob that it was lost by 2nd P.O.?

Condition

Sol $P(L) = \frac{1}{5}$ & $P(NL) = \frac{4}{5}$

Let we have Total (25) Parcels

$P(L \text{ by } 1^{st} P.O) = \frac{1}{5} = \frac{5}{25}$ & $P(NL \text{ by } 1^{st} P.O) = \frac{20}{25}$

$P(L \text{ by } 2^{nd} P.O) = \frac{1}{5} = \frac{4}{20}$ & $P(NL \text{ by } 2^{nd} P.O) = \frac{16}{20}$

$S \xrightarrow[25]{} (P.O\,1) \xrightarrow[=25]{\frac{4}{5}} (P.O\,2) \xrightarrow[(=20)]{\frac{4}{5}} R \quad (=16)$

original Prob = P(S) = 1

Reduced Prob = P(condition) = P(parcel is lost)

$$= P[\text{either lost by } 1^{st} \text{ or lost by } 2^{nd}]$$

(ME)

$$= P(\text{lost by } 1^{st}) + P(\text{lost by } 2^{nd})$$

$$= P(\text{lost by } 1^{st}) + P[(\text{NL by } 1^{st}) \& (\text{Lost by } 2^{nd})]$$

$$= \frac{1}{5} + \frac{4}{5} \times \frac{1}{5} = \frac{9}{25}$$

fav Prob = $P(\text{Lost by } 2^{nd} \text{ P.O}) = P[\text{NL by } 1^{st} \text{ \& Lost by } 2^{nd}] = \frac{4}{5} \times \frac{1}{5} = \frac{4}{25} = \frac{4/25}{1}$

So Conditional Prob = $\frac{\text{fav Prob}}{\text{R. Prob}} = \frac{4/25}{9/25} = \frac{4}{9}$

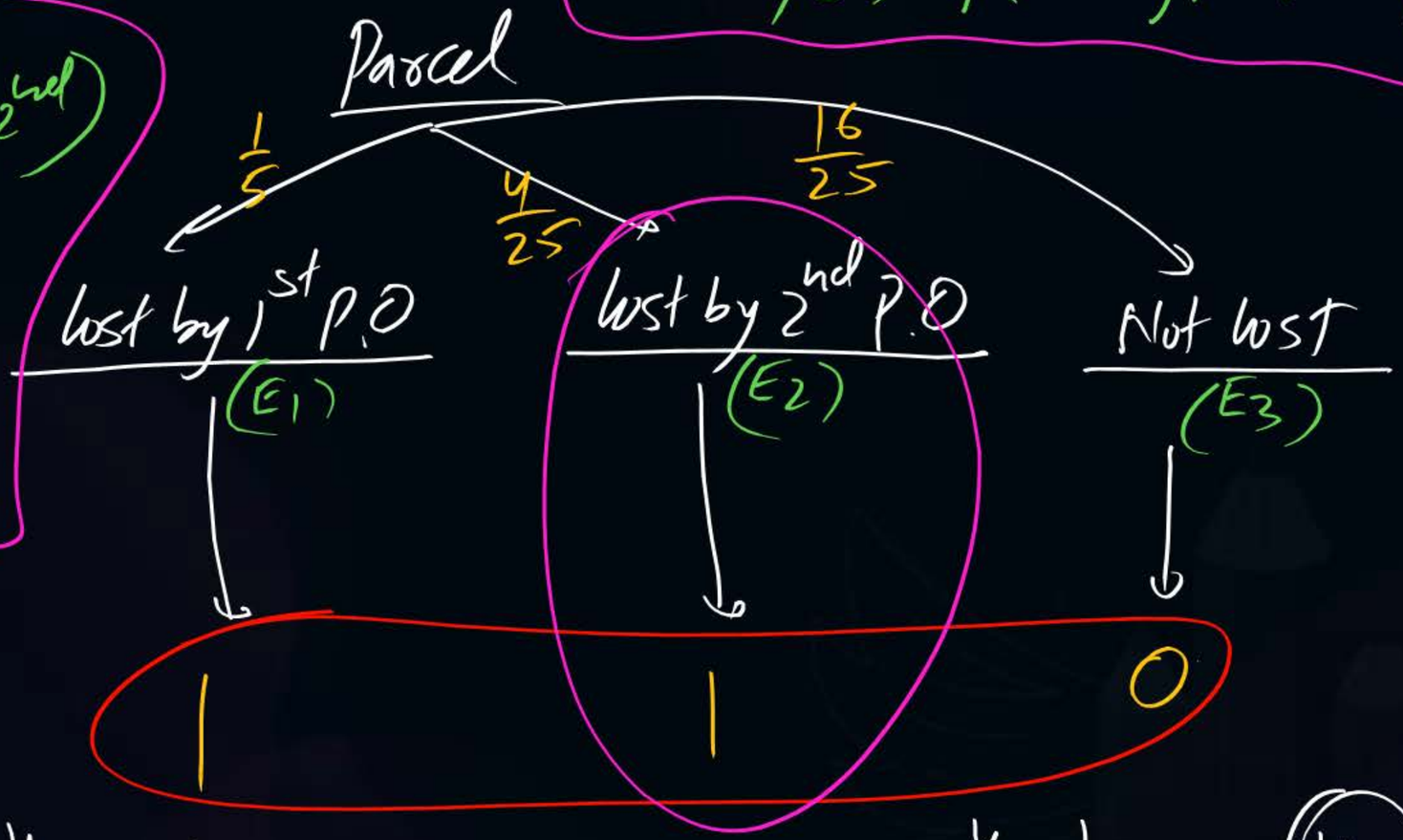
(M-II) Using B-Th.:- $A = \{\text{Parcel is lost}\}$, $P(\text{lost by } 1^{st}) = \frac{1}{5}$

$P(\text{lost by } 2^{nd}) = P(\text{NL by } 1^{st} \text{ \& L by } 2^{nd}) = \frac{4}{5} \times \frac{1}{5}$

$P(\text{NL by any one}) = P(\text{NL by } 1^{st} \text{ \& NL by } 2^{nd}) = \frac{4}{5} \times \frac{4}{5}$

$\because P(E_1) + P(E_2) + P(E_3) = 1$

i.e. Events are ME & Exh



Parcel is lost A:

$$P(A) = \frac{1}{5} \times 1 + \frac{4}{25} \times 1 + \frac{16}{25} \times 0 = \frac{9}{25} \quad \& \quad P(E_2/A) = \frac{\frac{4}{25} \times 1}{9/25} = \frac{4}{9}$$

11. Q. A coin is tossed twice then P(Both are H) = ? if we have <u>at least one Head</u>

Ans = $\frac{1}{3}$

original SSp = {(HH), (HT), (TH), (TT)} = 4

Reduced SSp = {At least one H} = {(HT), (TH), (HH)} = 3

fav Cases = {Both are H} = {(HH)} = 1

Cond^{n} Prob = $\frac{\text{fav Cases}}{\text{RSSp}} = \frac{1}{3}$

$S = \{(HH), (HT), (TH), (TT)\} = 4$

App (3) $A = \{\text{Both are } H\} = \{(HH)\} = 1 \Rightarrow P(A) = \frac{1}{4}$

$B = \{\text{At least one } H\} = \{(HH), (HT), (TH)\} = 3 \Rightarrow P(B) = \frac{3}{4}$

$A \cap B = \{(HH)\} = 1 \Rightarrow P(A \cap B) = \frac{1}{4} = \frac{1/4}{1}$

& Cond^{l} Prob $P(A/B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{3/4} = \frac{1}{3}$

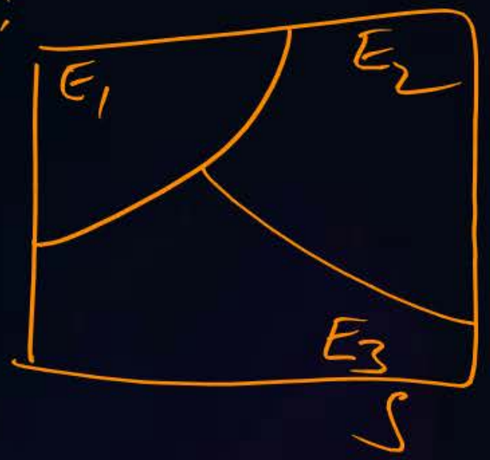
M.E Events → E_1, E_2, E_3 are called ME if $E_i \cap E_j = \phi \ \forall i \neq j$ $\Leftrightarrow$

$$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_2 \cap E_3) - P(E_3 \cap E_1) + P(E_1 \cap E_2 \cap E_3)$$

$$\boxed{P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3)} \quad -0-0-0+0$$

Exhaustive Events → E_1, E_2, E_3 are called Exhaustive if $\boxed{E_1 \cup E_2 \cup E_3 = S}$ $\Leftrightarrow$

Note:



By Diagram we can conclude that E_1, E_2, E_3 are ME & Exhaustive

Sp Note: if E_1, E_2, E_3 are ME & Exhaustive Events, then

$$\boxed{P(E_1) + P(E_2) + P(E_3) = 1}$$ learn.

Proof: $P(E_1 \cup E_2 \cup E_3) = P(S) \Rightarrow P(E_1) + P(E_2) + P(E_3) = 1$ Hence Proved

Eg: $S=\{1,2,3,4,5,6\}$ & let $E_1=\{1,2\}$, $E_2=\{3\}$, $E_3=\{4,5,6\}$

$\Rightarrow P(E_1)=\frac{2}{6}$, $P(E_2)=\frac{1}{6}$, $P(E_3)=\frac{3}{6}$

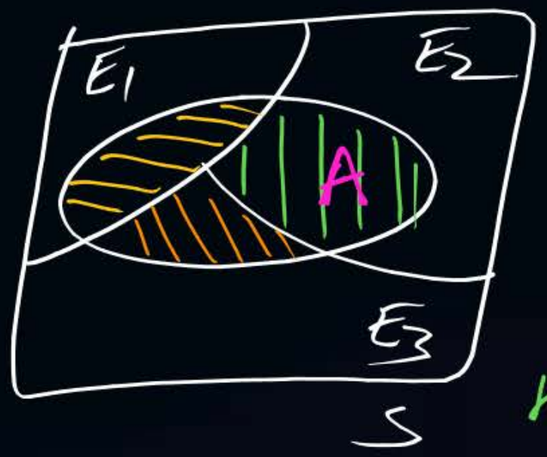
∵ $E_1 \cup E_2 \cup E_3 = S$ & E_1, E_2, E_3 are Exhaustive

∵ $E_1 \cap E_2 = E_2 \cap E_3 = E_3 \cap E_1 = \phi$ i.e. $E_i \cap E_j = \phi \ \forall i \neq j$ & these are ME also.

Hence $P(E_1)+P(E_2)+P(E_3) = \frac{2}{6}+\frac{1}{6}+\frac{3}{6} = \frac{6}{6} = 1$ Hence verified

Law of Total Probability

Let E_1, E_2, E_3 are ME and Exhaustive events, and A is an Event which can occur with all then



$$P(A) = P(E_1)P(A/E_1) + P(E_2)P(A/E_2) + P(E_3)P(A/E_3) \quad \text{---(1)}$$

Here E_1 & A are not ME
E_2 & A " " "
E_3 & A " " "

But E_1, E_2, E_3 are M.E

Baye's Th.e

theory same as above. then

$$P(E_1/A) = \frac{P(E_1)P(A/E_1)}{P(A)}, \quad P(E_2/A) = \frac{P(E_2)P(A/E_2)}{P(A)}, \quad P(E_3/A) = \frac{P(E_2)P(A/E_3)}{P(A)}$$

(Inverse Prob Th)

Note ① $P(A/E_1)$ = Prob of A, when E_1 has already occurred

$P(E_1/A)$ = Prob of E_1, when A has " "

② N. Cond^n for the existence of Baye's Th & Law of Total Prob is "Associated Events must be ME & Exhaustive"

③ In Baye's Th; $A = \{$ Assume that event as A which is given as Condition $\}$

& In Law of Total Prob, $A = \{$ Assume that Event at A which is Required $\}$

④ To solve tough Questions of Conditional Prob, we will use Baye's Th.

⑤ Whenever Question is Based on Conditional Prob & there is a feeling of Cross Check something then we should use B-Th.

⑥ If we have no Condition in Quest then we should use Law of Total Prob.

Q. In a town there are 70% Men & 30% Women in which 80% Men & 50% Women are employed. If a person is selected at Random then find the prob. that person is an unemployed person?

Let $E_1 = \{\text{Man}\} \Rightarrow P(E_1) = 0.7$

$E_2 = \{\text{Woman}\} \Rightarrow P(E_2) = 0.3$

$\Rightarrow E_1$ & E_2 are M.E

E_1 & E_2 are also exhaustive

$\therefore P(E_1) + P(E_2) = 1$

Let $A = \{\text{unemployed Person}\}$

& $P(A/E_1) = P(\text{unemployed Man}) = 20\% = 0.2$

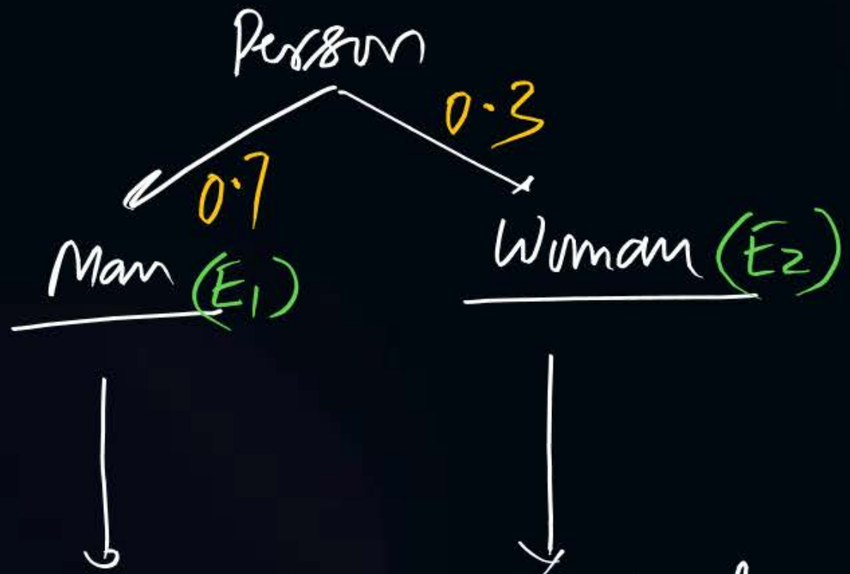
$P(A/E_2) = P(\text{unemployed Woman}) = 50\% = 0.5$

By Law of Total Prob. :-

$$P(A) = P(E_1)P(A/E_1) + P(E_2)P(A/E_2)$$

$$= (0.7 \times 0.2) + (0.3 \times 0.5) = 0.29 \simeq 29\%$$

(M-II) let $A = \{$ unemployed Person $\}$



E_1 & E_2 are (ME) as well as Exhaustive

$\because P(E_1) + P(E_2) = 1$

$$P(A) = (0.7 \times 0.2) + (0.3 \times 0.5)$$

$$= 0.29 \simeq 29\%$$

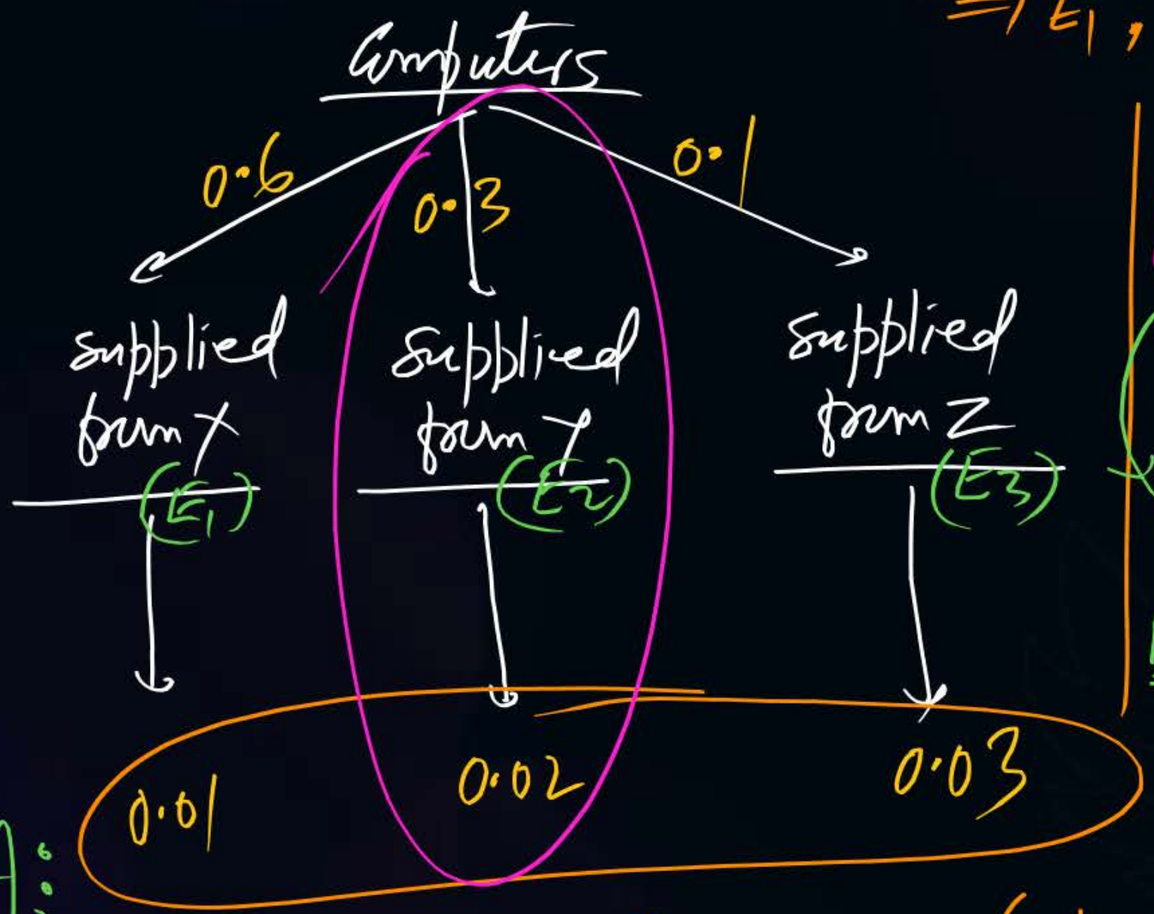
Qe Computers are supplied to an organisation according to the following chart;

Company	% of computers supplied	Prob of being Defective
X	60%	0.01
Y	30%	0.02
Z	10%	0.03

(i) Find the Prob that supplied computer is defective?

(ii) Given that, supplied Computer is defective (A) then find the Prob that it was supplied from Company Y? (Condition)

$A = \{$ supplied Comp is defective $\}$



Supplied Comp is Defective A:

$P(A) = (0.6 \times 0.01) + (0.3 \times 0.02) + (0.1 \times 0.03) = 0.015 = \frac{15}{1000}$

$\because P(E_1) + P(E_2) + P(E_3) = 1$

$\Rightarrow E_1, E_2, E_3$ are ME & Exhaustive.

$$P(E_2/A) = \frac{P(\text{V. Path})}{P(\text{H. Path})}$$

$$= \frac{0.3 \times 0.02}{0.015} = \frac{6}{15}$$

Note: Cond^n Prob $= \frac{\text{fav Prob}}{\text{R-Prob}} = \frac{0.006}{0.015}$

Note :- (1) $P(A) = 0.015 = \frac{15}{1000}$, $P(E_2/A) = \frac{6}{15}$

Similarly $P(E_1/A) = \frac{0.6 \times 0.01}{0.015} = \frac{6}{15}$ & $P(E_3/A) = \frac{0.1 \times 0.03}{0.015} = \frac{3}{15}$

(2) Analysis :- out of 1000 supplied Computers only 15 are Defective

4 out of 15 Defective Computers, 6 are supplied from X, 6 from Y & 3 from Z

Q: In a test, an Examinee either Guesses or Copy or know the answer of an MCQ (≡ 4 choices)

· The prob of making Guess is $\frac{1}{3}$ & Copying is $\frac{1}{6}$.

The prob that his answer is Correct given that he Copied it is $\frac{1}{2}$

then find the prob that he copy the answer if he answered Correctly?

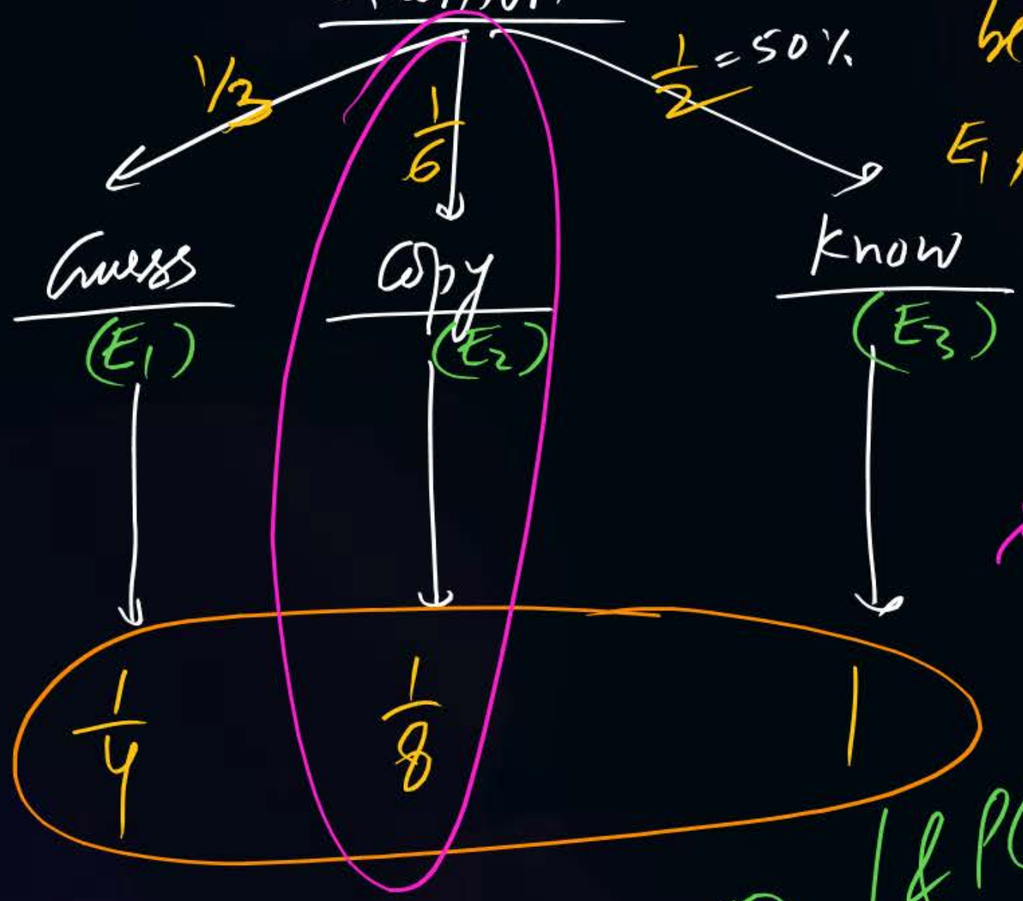
Condition

Sol: $A = \{$ Examinee answered Correctly $\}$

$E_1 = \{$ Guess Method $\}$

$E_2 = \{$ Copy $\}$

$E_3 = \{$ knows $\}$



E_1, E_2, E_3 are ME ∵ Single & Can not be solved any two methods at a time

E_1, E_2, E_3 are also Exhaustive ∵ we have no 4th method & Examinee has to solve all the Ques by these 3 methods

$= 100\%$

So $P(E_1)+P(E_2)+P(E_3)=1$

$\frac{1}{3}+\frac{1}{6}+P(E_3)=1 \Rightarrow P(E_3)=\frac{1}{2}$

Examinee answered correctly

A:

$P(A)=\left(\frac{1}{3}\times\frac{1}{4}\right)+\left(\frac{1}{6}\times\frac{1}{8}\right)+\left(\frac{1}{2}\times 1\right)=\frac{29}{48}$

& $P(E_2/A)=\frac{\frac{1}{6}\times\frac{1}{8}}{29/48}=\frac{1}{29}$

Analysis :- ① $P(A)=\frac{29}{48}$, $P(E_2/A)=\frac{1}{29}$, $P(E_1/A)=\frac{4}{29}$, $P(E_3/A)=\frac{24}{29}$

② out of 48 Total questions, 29 are Correct answers and out of 29 Correct answers, he Guessed 4, Copied 1 and he knew 24

③ Total Choices of Selecting 1 option out of 4 options $= {}^4C_1$

fav " " 1 correct option $= {}^1C_1$

So $P(\text{Correct Ans / Guess Method}) = P(A/E_1) = \frac{\text{fav}}{\text{Total}} = \frac{{}^1C_1}{{}^4C_1} = \boxed{\frac{1}{4}}$

HW8 A person is known to speak truth 3 out of 4 times. He throw a die and Reports that it is six then find the prob that it is actually six?

Ans = $\frac{3}{8}$

M-I →

M-II →

DOUBT

$P(G_1) = P(G_2) = P(G_3) = P(G_4) = 0.2$, Ind

M-I $P(G_2 \cap G_3 / G_1) = ? = P(G_2 \cap G_3) = 0.2 \times 0.2 = 0.04$

THANK - YOU

M-II Req Prob $= P[G_1 \cap G_2 \cap G_3 \cap$ (something occurs)$]$

$= 1 \times 0.2 \times 0.2 \times 1 = 0.04$